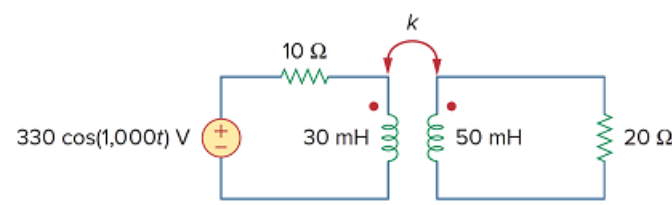


Problem

In the circuit of Fig. 13.98, find the value of the coupling coefficient k that will make the $10\text{-}\Omega$ resistor dissipate 1.28 kW . For this value of k , find the energy stored in the coupled coils at $t = 1.5\text{ s}$.

Figure 13.98



Step-by-step solution

Step 1 of 21

Refer to Figure 13.98 from the textbook.

Write the value of the source voltage, V_1 .

$$V_1 = 330 \cos(1,000t) \text{ V} \dots\dots (1)$$

$$V_1 = 330 \angle 0^\circ$$

Write the standard equation for a voltage source.

$$v = V_m \cos \omega t \dots\dots (2)$$

Compare equations (1) and (2) and obtain the value of ω .

$$\omega = 1,000 \text{ rad/s}$$

Step 2 of 21

Assume the direction of the loop currents I_1 and I_2 as shown in Figure 1.

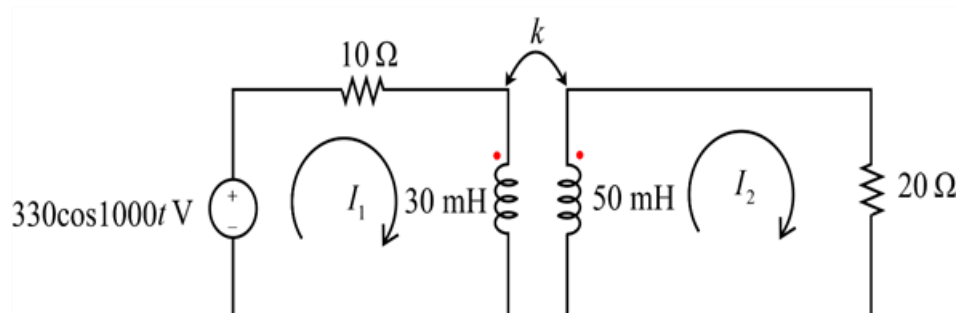


Figure 1

Step 3 of 21

Calculate the reactance of the primary winding (L_1) of the transformer.

$X_{L_1} = j\omega L_1$ Substitute 30 mH L_1 and 1,000 rad/s for ω .

$$\begin{aligned} X_{L_1} &= j(1,000)(30 \times 10^{-3}) \\ &= j30 \Omega \end{aligned}$$

Step 4 of 21

Calculate the reactance of the secondary winding (L_2) of the transformer.

$X_{L_2} = j\omega L_2$ Substitute 50 mH L_2 and 1,000 rad/s for ω .

$$\begin{aligned} X_{L_2} &= j(1,000)(50 \times 10^{-3}) \\ &= j50 \Omega \end{aligned}$$

Step 5 of 21

The magnitude of current I_1 gives its maximum value.

Calculate the rms value of the current I_1 .

$$I_{1rms} = \frac{|I_1|}{\sqrt{2}}$$

Write the expression for the power, p dissipated in a resistor, R_1 .

$$p = I_{1rms}^2 R_1$$

$$\begin{aligned} p &= \left(\frac{|I_1|}{\sqrt{2}} \right)^2 R_1 \\ &= \frac{1}{2} |I_1|^2 R_1 \end{aligned}$$

Substitute 1.28 kW for p and 10 Ω for R_1 .

$$1.28 \times 10^3 = \frac{1}{2} |I_1|^2 (10)$$

$$|I_1|^2 = \frac{(1.28 \times 10^3)(2)}{10}$$

$$|I_1|^2 = 256$$

$$|I_1| = 16 \text{ A}$$

Step 6 of 21

Calculate the magnitude of the input impedance, $|Z_{in}|$.

$$Z_{in} = \frac{V_1}{I_1}$$

$$|Z_{in}| = \frac{|V_1|}{|I_1|}$$

Substitute 330 V for $|V_1|$ and 16 A for $|I_1|$.

$$Z_{in} = \frac{330}{16}$$

$$= \frac{300}{16}$$

$$20.625 \Omega$$

Step 7 of 21

Write the expression for the reflected impedance, Z_R reflected to the primary side.

$$Z_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L} \dots\dots (3)$$

Here,

M is the mutual inductance between the two coils,

R_2 is the resistance on the secondary side of the transformer,

Z_L is the load impedance.

Since no load impedance is present in the circuit.

Thus,

Substitute 0 for Z_L in equation (3).

$$Z_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2} \dots\dots (4)$$

Substitute 1000 rad/s for ω , 20 Ω for R_2 and 50 mH for L_2 in equation (4).

$$Z_R = \frac{(1000)^2 M^2}{20 + j(1000)(50 \times 10^{-3})}$$

$$= \left(\frac{M^2}{20 + j50} \right) \times 10^6$$

Step 8 of 21

Write the expression for the input impedance, Z_{in} .

$$Z_{in} = R_1 + j\omega L_1 + Z_R$$

Here, R_1 is the resistance of the primary side of the transformer.

Substitute 1000 rad/s for ω , 10 Ω for R_1 , 30 mH for L_1 and $\left(\frac{M^2}{20 + j50} \right) \times 10^6$ for Z_R .

$$Z_{in} = 10 + j(1000)(30 \times 10^{-3}) + \left(\frac{M^2}{20 + j50} \right) \times 10^6$$

$$= 10 + j30 + \left(\frac{M^2}{20 + j50} \right) \times 10^6$$

$$= \frac{(10^6 M^2 - 1300) + j1100}{20 + j50}$$

Calculate the magnitude of Z_{in} .

$$|Z_{in}| = \frac{\left(\sqrt{(10^6 M^2 - 1300)^2 + (1100)^2} \right)}{\sqrt{(20)^2 + (50)^2}}$$

$$= \frac{\left(\sqrt{(10^6 M^2 - 1300)^2 + 12100} \right)}{\sqrt{2900}}$$

Substitute 20.625Ω for Z_{in} .

$$20.625 = \frac{\left(\sqrt{(10^6 M^2 - 1300)^2 + 121 \times 10^4} \right)}{\sqrt{2900}}$$

Step 10 of 21

Square both the sides.

$$(20.625)^2 = \left(\frac{(10^6 M^2 - 1300)^2 + 121 \times 10^4}{2900} \right)$$

$$(425.39)(2900) = \left((10^6 M^2 - 1300)^2 + 121 \times 10^4 \right)$$

$$1,233,632.813 - 1,210,000 = (10^6 M^2 - 1300)^2$$

[Comments \(1\)](#)



Anonymous

a million ? lmao

Step 11 of 21

Take square root on both the sides.

$$\sqrt{23632.8125} = (10^6 M^2 - 1300)$$

$$153.73 = (10^6 M^2 - 1300)$$

$$153.73 + 1300 = 10^6 M^2$$

$$M^2 = 1453.73 \times 10^{-6}$$

$$M = \sqrt{1453.73 \times 10^{-6}}$$

$$M = 38.128 \times 10^{-3} \text{ H}$$

$$M = 38.128 \text{ mH}$$

Step 12 of 21

Write the expression for the coefficient of coupling, k .

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

Substitute 38.128 mH for M , 30 mH for L_1 and 50 mH for L_2 .

$$\begin{aligned} k &= \frac{38.128}{\sqrt{(30)(50)}} \\ &= 0.9845 \end{aligned}$$

Therefore, the coefficient of coupling, k is 0.9845.

Step 13 of 21

Calculate the reactance of mutual inductance, X_M .

$$X_M = j\omega M$$

Substitute 38.128 mH for M .

$$\begin{aligned} X_M &= j(1000)(38.128 \times 10^{-3}) \\ &= j38.128 \Omega \end{aligned}$$

Step 14 of 21

Modify the circuit by replacing the coil with its respective reactance as shown in Figure 2.

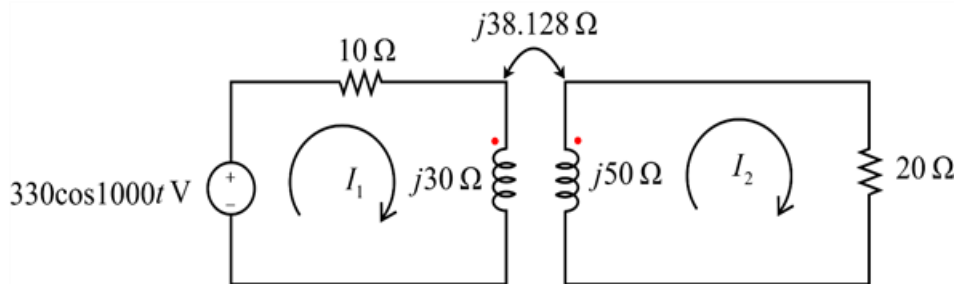


Figure 2

Step 15 of 21

Apply dot convention to determine the polarities of the mutual voltages. If the current enters at the dot of one coil, then the polarity of the mutual voltage is positive at the dot in the other coil.

Since the current is entering at the dot of L_1 , the polarity of the mutual voltage of L_2 is positive at the dot.

Since the current is leaving at the dot of L_2 , the mutual voltage in L_1 is negative at the dot.

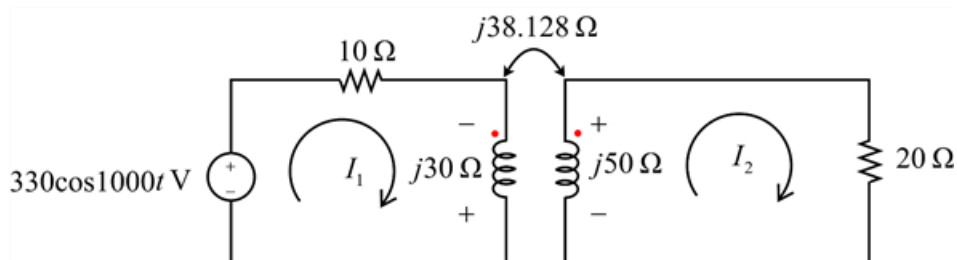


Figure 3

Step 16 of 21

Observe the polarities of the mutual voltages in L_1 and L_2 . The two coils are in series-opposition with each other.

Apply Kirchhoff's Voltage law in loop with loop current I_1 .

$$330 = (10 + j30)I_1 - j38.128I_2 \dots\dots (5)$$

Apply Kirchhoff's Voltage law in loop with loop current I_2 .

$$0 = (20 + j50)I_2 - j38.128I_1$$

$$I_2 = \frac{j38.128}{(20 + j50)}I_1 \dots\dots (6)$$

Substitute $\frac{j38.128}{(20 + j50)}I_1$ for I_2 in equation (5).

$$330 = (10 + j30)I_1 - (j38.128)\left(\frac{j38.128}{(20 + j50)}I_1\right)$$

$$330 = (10 + j30)I_1 + \frac{1453.744}{(20 + j50)}I_1$$

$$330 = \frac{-1300 + j1100 + 1453.744}{(20 + j50)}I_1$$

$$330 = \frac{153.744 + j1100}{20 + j50}I_1$$

Step 17 of 21

Calculate I_1

$$\begin{aligned} I_1 &= 330 \left(\frac{20 + j50}{153.744 + j1100} \right) \\ &= 330(-0.047 - j0.012) \\ &= 15.535 - j3.829 \\ &= 16 \angle -13.84^\circ \text{ A} \end{aligned}$$

Step 18 of 21

Calculate the instantaneous current, i_1 .

$$i_1 = 16 \cos(1000t - 13.84^\circ) \text{ A} \dots\dots (7)$$

At $t = 1.5 \text{ ms}$

$$\begin{aligned} 1000t &= (1000)(1.5 \times 10^{-3}) \\ &= 1.5 \text{ rad} \\ &= 85.94^\circ \end{aligned}$$

Substitute 85.94° for $1000t$ in equation (7).

$$\begin{aligned} i_1 &= 16 \cos(85.94 - 13.84^\circ) \\ &= 16(0.3074) \\ &= 4.918 \text{ A} \end{aligned}$$

[Comments \(1\)](#)



Anonymous

t= 1.5seconds not ms

Step 19 of 21

Substitute $16\angle -13.84^\circ \text{ A}$ for I_1 in equation (6).

$$\begin{aligned} I_2 &= \frac{j38.128}{(20 + j50)}(16\angle -13.84^\circ) \\ &= (0.657 + j0.263)(16\angle -13.84^\circ) \\ &= 11.2192 + j1.5685 \\ &= 11.328\angle 7.966^\circ \text{ A} \end{aligned}$$

Step 20 of 21

Calculate the instantaneous current, i_2 .

$$i_2 = 11.328 \cos(1000t + 7.966^\circ) \text{ A}$$

Substitute 85.94° for $1000t$.

$$\begin{aligned} i_2 &= 11.328 \cos(85.94 + 7.966^\circ) \\ &= 11.328(-0.068) \\ &= -0.77 \text{ A} \end{aligned}$$

Step 21 of 21

Write the expression for the energy stored in the coupled coils in series opposition.

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 - M i_1 i_2$$

Substitute 4.918 A for i_1 , -0.77 A for i_2 , 30 mH for L_1 , 50 mH for L_2 and 38.128 mH for M .

$$\begin{aligned} w &= \frac{1}{2}(30)(4.918)^2 + \frac{1}{2}(50)(-0.77)^2 - (38.128)(4.918)(-0.77) \\ &= \frac{1}{2}(30)(24.186) + \frac{1}{2}(50)(0.5929) + 144.39 \\ &= 362.79 + 14.8225 + 144.39 \\ &= 522 \text{ mJ} \end{aligned}$$

Therefore, the energy stored in the couple coils, w is **522 mJ**.